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Read the following passage.

Complete achromatopsia is a form of complete colour blindness. It is caused by having only rods and no functional cone cells. People with complete achromatopsia have difficulty in seeing detail. Complete achromatopsia is caused by an autosomal recessive allele and is usually very rare in populations with only one in 40 000 being affected. However on the Pacific island of Pingelap ten percent of the population are affected.

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One form of red-green colour blindness is caused by a sex-linked recessive allele which affects more men than women. People with this red-green colour blindness are unable to distinguish between red and green, and also between other colours. They have green-sensitive cones but the photoreceptive pigment they contain does not function.

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Scientists investigated the use of gene therapy to correct red-green colour blindness in monkeys. They injected viruses containing the gene for the green-sensitive pigment directly into the eyes of the monkeys. Although the monkeys maintained two years of colour vision, there is debate on whether this form of gene therapy is worthwhile. No clinical trials of this procedure have been carried out on humans. Current research into the treatment of red-green colour blindness involves the use of induced pluripotent stem cells (iPS cells). The use of iPS cells could have advantages over the use of gene therapy.

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Use the information in the passage and your own knowledge to answer the following questions.

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People with complete achromatopsia have difficulty in seeing detail (lines 2–3).

Explain why.

[3 marks]

1 0 . 2

Ten percent of the population on the Pacific island of Pingelap are affected by complete achromatopsia (lines 3–6).

Use the Hardy-Weinberg equation to calculate the percentage of this population who are heterozygous for this disorder.

Show your working.

[2 marks]

Answer = _____ %

1 0 . 3

Red-green colour blindness affects more men than women (lines 7–8).

Explain why.

[2 marks]

Question 10 continues on the next page

1 0 . 4

People with red-green colour blindness are unable to distinguish between red and green, and also between other colours (lines 8–10).

Explain why.

[3 marks]

1 0 . 5

Current research into the treatment of red-green colour blindness involves the use of induced pluripotent stem cells (iPS cells) (lines 17–19).

Suggest how iPS cells could correct red-green colour blindness.

[2 marks]

The use of iPS cells could have advantages over the use of gene therapy to correct red-green colour blindness (lines 19–20).

[3 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

END OF QUESTIONS